IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

GAME THEORY FOR MULTI-AGENT DECISION MAKING

Monday, 18 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer FOUR questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : R. Vinter
Second Marker(s) : Y. Gao

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MSc and EEE Part IV: M.Eng. and ACGI

ELEC70136: GAME THEORY FOR MULTI-AGENT DECISION MAKING

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Information for candidates:
The Finite Horizon LQ Control Problem

$$(P_{0,x_0}) \begin{cases} \text{Minimise } \sum_{t=0}^{N-1}(x_t^T Q x_t + u_t^T u_t) + x_N^T G x_N, \\ \text{over controls } u_0, \ldots, u_{N-1} \text{ and states } x_0, \ldots, x_N \text{ satisfying} \\ x_{t+1} = Ax_t + Bu_t \quad \text{for } t = 0, \ldots, N-1, \\ x_0 \text{ given,} \end{cases}$$

in which N is an integer, A, Q and G are $n \times n$ matrices (Q and G are symmetric and non-negative definite), B is an $n \times m$ matrix.

The optimal control $\{\bar{u}_t\}$ and state $\{\bar{x}_t\}$ satisfy

$$\bar{u}_t = -(I + B^T P_{t+1} B)^{-1} B^T P_{t+1} A \bar{x}_t \quad \text{for } t = 1, \ldots, N-1, \tag{1}$$

in which $P_0, \ldots, P_N$ are symmetric, non-negative definite matrices, obtained from the recursion:

$$\begin{cases} P_t = Q + A^T P_{t+1} A - A^T P_{t+1} B (I + B^T P_{t+1} B)^{-1} B^T P_{t+1} A \quad \text{for } t = N-1, \ldots, 0 \\ P_N = G\,. \end{cases} \tag{2}$$

The minimum cost is $x_0^T P_0 x_0$.

1: Two players, P_1 and P_2, compete in a non-zero sum matrix game, using mixed strategies. Both players seek to minimise their pay-off functions, which are

$$J^1(x,y) = x^T A y \text{ and } J^2(x,y) = x^T B y,$$

for given mixed strategies $x \in \Lambda^{(n)}$ and $y \in \Lambda^{(m)}$ of P_1 and P_2. Here, A and B are given $n \times m$ matrices and $\Lambda^{(r)}$ is the set of probability mass functions on R^r:

$$\Lambda^{(r)} := \{p = [p_1, \ldots, p_r]^T \in R^r \, : \, p_i \geq 0 \text{ for all } i \text{ and } \sum_i p_i = 1\}.$$

(i): Assume $A = \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 \\ 4 & 1 \end{bmatrix}$.

(1): Use graphical techniques to find a pair of mixed strategies $((1-\bar{\lambda}, \bar{\lambda}), (1-\bar{\mu}, \bar{\mu}))$ for which $(1-\bar{\lambda}, \bar{\lambda})$ is a mixed safety strategy for player P_1 and $(1-\bar{\mu}, \bar{\mu})$ is a mixed safety strategy for player P_2. [6 marks]

Show that $((1-\bar{\lambda}, \bar{\lambda}), (1-\bar{\mu}, \bar{\mu}))$ is not a Nash Equilibrium. [2 marks]

(2): The above game has a Nash equilibrium $((1-\lambda', \lambda'), (1-\mu', \mu'))$ in mixed strategies, which is strictly mixed, in the sense

$$0 < \lambda' < 1 \text{ and } 0 < \mu' < 1\,.$$

Use analytical techniques to determine $((1-\lambda', \lambda'), (1-\mu', \mu'))$. [6 marks]

Comment on the fact that, for the above choice of pay-off matrices, the safety and Nash equlibrium strategies do not coincide. [2 marks]

(ii): Suppose that $\bar{x} \in \Lambda^{(n)}$ is a safety strategy for P_1. Let $\bar{p}$ be the number

$$\bar{p} := \max_{y \in \Lambda^{(m)}} \bar{x}^T A y.$$

Show that $(\bar{p}, \bar{x})$ is also a solution of the optimisation problem

$$(L) \begin{cases} \text{Minimise } p \\ \text{over } p \in R \text{ and } x = [x_1, \ldots, x_n]^T \in R^n \text{ such that} \\ p\, 1^{(n)} \geq x^T A\,. \\ x_i \geq 0 \text{ for } i = 1, \ldots, n \text{ and } 1^{(n)} x = 1, \end{cases}$$

in which $1^{(n)}$ is the row n-vector $[1, \ldots, 1]$. [4 marks]

Hint: show that, for any $y \in \Lambda^{(m)}$ and pair (p, x) satisfying the constraints of (L),

$$p \geq x^T A y.$$

.

2: Consider a 2-stage zero-sum game in intrinsic form, involving players A and B, who seek to maximise and minimise their pay-offs, respectively. See the diagram in Figure 1.

Key Features

- A and B choose from $\{L, R\}$ ('left' and 'right') in both stages of the game.
- When A chooses a strategy in stage 2, A knows B's strategy in Stage 1. There is no transfer of information from A to B. (The information sets involved are illustrated in the diagram.)

Reformulate this game as a normal form zero-sum game. You will need to:

(i): Identify the strategy spaces associated with players A and B, containing 8 and 4 strategies respectively, to take account of the information sets involved. [4 marks]

Hint: A strategy for A is specified, for example, as 'in stage 1, A chooses L, and in stage 2, A chooses R if B has chosen L (in stage 1) and R if B chosen R .

(ii): Construct the pay-off matrix for the normal form game. [12 marks]

(iii): Identify any Nash equilibria in pure strategies, evaluate the pay-offs and state whether they are admissible. [4 marks]

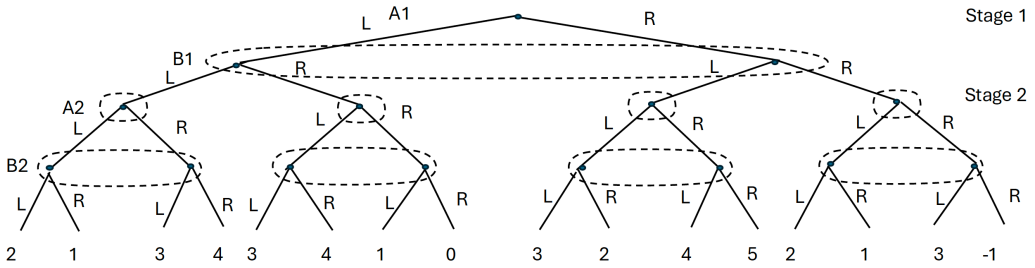


Figure 1: Two Stage, Zero Sum Game in Extensive Form.

3: Two companies C_1 and C_2, who make the same product, seek to maximise their profits by choosing the best prices for their products.

Denote by d_1 the demand for products from C_1, i.e., the number of items that C_1 can sell. Denote by d_2 the demand for products from company C_2.

Demand depends on the prices p_1 and p_2 set by the two companies, according to the Edgeworth pricing law:

$$d_1 = 4 - 2p_1 - p_2 \text{ and } d_2 = 4 - 2p_2 - p_1$$

Assume that each company produces exactly enough items to satisfy demand. Then their profits P_1 and P_2 are

$$P_1 = d_1 \times p_1 - 0.5 \times (d_1)^2$$
$$P_2 = d_2 \times p_2 - 0.25 \times (d_2)^2$$

('profit from sales minus production cost')

(i): Derive formulae for the pay-offs $J_1(p_1, p_2)$ and $J_2(p_1, p_2)$ of the two companies. [2 marks]

(ii): Determine the Nash equilibrium $(\bar{p}_1, \bar{p}_2)$ for the prices. [8 marks]

(iii): Sketch the reaction curves for the two companies. Show that they are consistent with the Nash equilibrium calculated in part (ii). [5 marks]

(iv): Take Company C_1 as the leader. Determine the Stackelberg optimising price for Company C_1 and the corresponding price for Company C_2. [5 marks]

4: For an n-dimensional evolutionary game the fitness functions are

$$(a_1(x), \ldots, a_n(x)),\ x \in \Lambda^{(n)},$$

in which $\Lambda^{(n)} := \{(x_1, \ldots, x_n) :\ x_1, \ldots, x_n \geq 0 \text{ and } \sum_{i=1}^{n} x_i = 1\}$.
(Members of each sub-population seek to maximise their fitness.)

(i): Give the definitions of an (evolutionary) equilibrium and of a stable equilibrium. [2 marks]

(ii): Suppose that, for some $n \times n$ matrix A,

$$\begin{bmatrix} a_1(x) \\ \vdots \\ a_n(x) \end{bmatrix} = Ax, \quad \text{for } x \in R^n$$

i.e., the game is a matrix evolutionary game.
Let $\bar{x}$ be an evolutionary equilibrium. Assume that

$$(\bar{x} - x)^T A(\bar{x} - x) < 0, \quad \text{for all } x \in \Lambda^n,\ x \neq \bar{x}\,.$$

Show that $\bar{x}$ is a stable equilibrium. [6 marks]

(iii): Take $n = 4$ and let the fitness functions be

$$\begin{aligned} a_1(x) &= 10 - x_1 - k_0(x_2 + x_3 + x_4) \\ a_2(x) &= 10 - x_2 - k_1 x_1 \\ a_3(x) &= 10 - x_3 - k_1 x_1 \\ a_4(x) &= 10 - x_4 - k_1 x_1\,, \end{aligned}$$

in which $k_0 > 0$ and $k_1 > 0$ are the constants

$$k_0 = 1/12 \text{ and } k_1 = 3/4\,.$$

Find a strictly interior Nash equilibrium $\bar{x} = (\bar{x}_1, \ldots, \bar{x}_n)$. [8 marks]

Hints: the equilibrium strategies are unaffected if the same constant is subtracted from each pay-off function. Use the fact that, due to symmetry, $\bar{x}_2 = \bar{x}_3 = \bar{x}_4$.

By using the results of part (ii), or otherwise, state a condition to test the stability of this equilibrium. [4 marks]

5(i): For a given number a consider the optimal control problem

$$(P)\begin{cases}\text{Minimise } J_a(\{u_i\}) = x_N^2 + \sum_{t=0}^{N-1} u_t^2 \\ \text{over controls } \{u_0, \ldots, u_{N-1}\} \text{ s.t.} \\ x_0 = 0,\end{cases}$$

in which the numbers $\{x_0, \ldots, x_N\}$ are calculated from $x(0) = 0$ and

$$x_{t+1} = x_t + a + u_t, \text{ for } t = 1, \ldots, N\,.$$

By considering a 2-vector state variable $z_t = [x_t, a]^T$, show that, if $\{\bar{u}_t\}$ is a minimiser for (P), it is also a minimiser for the standard Linear Quadratic (LQ) problem

$$(R)\begin{cases}\text{Minimise } \tilde{J}_a(z_0, \{u_i\}) = z_N^T G z_N + \sum_{t=0}^{N-1} u_t^2 \\ \text{over controls } \{u_0, \ldots, u_{N-1}\} \text{ s.t.} \\ z_{t+1} = Az_t + bu_t, \text{ for } t = 0, \ldots, N-1\,. \\ z(0) = (0, a),\end{cases}$$

in which the 2-dimensional vectors $\{z_0, \ldots, z_N\}$ are calculated from $z(0) = (0, a)$ and

$$z_{t+1} = Az_t + bu_t, \text{ for } t = 1, \ldots, N\,,$$

and evaluate the 2×2 matrices A and G and the 2-dimensional row vector b. [3 marks]

Hint: the second component of the state a satisfies $a_t = a_{t-1}$, because it is constant.

By quoting the formulae for the solution to the standard (LQ) problem (R), show that there exists a number d and sequences of numbers

$$\{\alpha_0, \ldots, \alpha_{N-1}\}, \{\beta_0, \ldots, \beta_{N-1}\}$$

and a sequence of 2×2 matrices $\{P_0, \ldots, P_N\}$ such the optimal controls $\bar{u}_0, \ldots, \bar{u}_{N-1}$ and state trajectories $\bar{x}_0, \ldots, \bar{x}_N$ for the original problem (P) satisfy

$$\bar{u}_t = -(\alpha_t \bar{x}_t + \beta_t a) \text{ for } t = 0, \ldots, N-1$$

and the minimum cost is $d \times a^2$.

Derive equations for the sequences $\{\alpha_t\}$, $\{\beta_t\}$ and $\{P_t\}$ and d. (You do not have to solve these equations.) [14 marks]

5(ii): Consider a Stackelberg game related to Problem (P), in which the leader chooses the number a and the follower chooses the control sequence $\{u_0, \ldots, u_{N-1}\}$. The pay-off functions, which the players seek to minimise, are

$$\begin{cases} L_a^{\text{follow}}(\{u_0, \ldots, u_{N-1}\}) = J_a(\{u_0, \ldots, u_{N-1}\})\,. \\ L_a^{\text{lead}}(\{u_0, \ldots, u_{N-1}\}) = J_a(\{u_0, \ldots, u_{N-1}\}) - 3\,a\,. \end{cases}$$

Using the results of part (i), or otherwise, derive formulae for the leader's optimising Stackelberg strategy a. [3 marks]